Linked List

In this exercise, you will implement a **music playlist** for a music player app. A key feature of such an app is that when the **last song finishes**, the playlist **automatically loops back** to the **first song** — meaning the playlist is **circular**.

You are provided the following node definition:

```
class AudioNode{
private:
    string title;
    Audio audio; // Assume Audio datatype exists, and is already defined
public:
    AudioNode * next;
    AudioNode * prev;
    AudioNode (string s, Audio A) : title(s), Audio(A), next(nullptr), prev(nullptr) {}
};
```

Q1 Provide the class definition for the class **Playlist** that manages these AudioNode as a doubly circular linked list. Use only one pointer in the class — head. **Do not** use a tail pointer or any additional list-level pointer. [Point(s): 1]

Q2 Now implement the Destructor. [Point(s): 2]

Q3 Now implement display() method, which prints the title of the songs. [Point(s): 2]

Operator Overloading

Q4 Now overload the operators + and = such that the following code works. [Point(s): 3]

Playlist NewPlaylist = OldPlaylist + AnAudio

Where OldPlaylist is an instance of the class Playlist and AnAudio is an instance of the class AduioNode (1 point for +, 2 points for =)

Inheritance

AudioNode is too general, now let's define specialized classes from AudioNode using inheritance.

Q5 For this task draw the UML diagrams for the following classes along with their relationship. [Point(s): 1]

- A new class class PodcastEpisode, which is an AudioNode, with the private data "PodcastName" of type string.
- A new class called VideoNode, with the private data "video" of datatype Video (assume there is such a data type)
- A new class called MusicVideo which is an AudioNode as well as a VideoNode.

Q6 Define constructor for MusicVideo. [Point(s): 1]

Bonus

Q7 What is wrong with the implementation of the Playlist class as done under the instructions of this quiz? [Point(s): 1]

Q8 Without overloading the operators for PodcastEpisode class, will this code work: NewPlaylist = OldPlaylist + APodcast? Explain your answer [Point(s): 1]